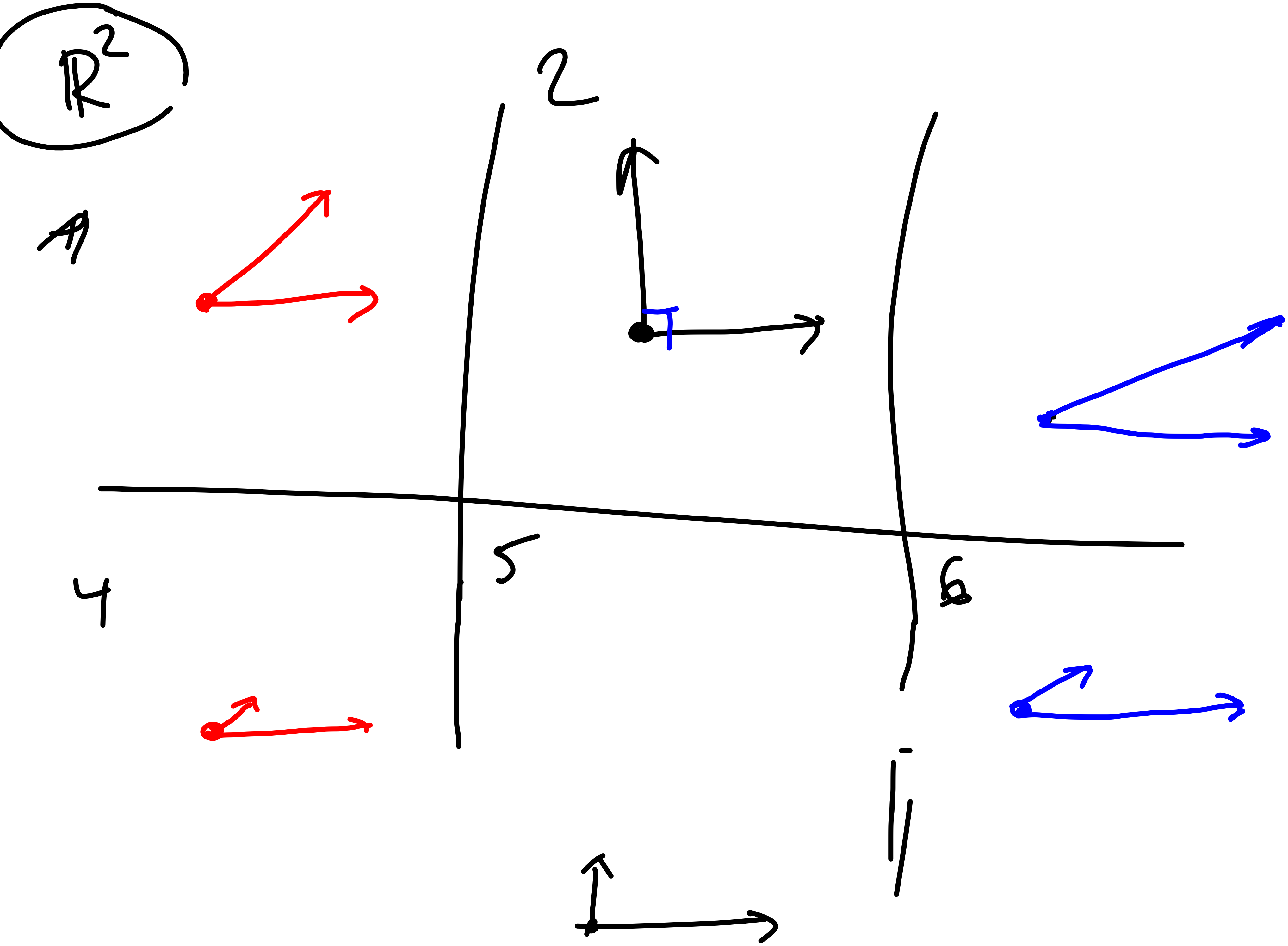


$\mathbb{R}^2$
2
4
5
6

Indreprodukt og orthogonalitet

Et sett med vektorer

- enten lin avh eller lin uavh
- kan vi kvantifisere lin avh?

(dvs mer/mindre lin avh/uavh)

$\vec{x} \in \mathbb{C}^n$, $\vec{y} \in \mathbb{C}^n$

Indre produkt: $(\vec{x}, \vec{y}) = \sum_{i=1}^{n} x_i \bar{y}_i = \vec{x} \cdot \vec{y}$

eller $(\vec{x}, \vec{y}) = \vec{y}^H \vec{x}$

($A^H = \overline{A^T}$, dvs transponert og kompl. konjugert)

Ex: $x = \begin{bmatrix} 2 & i & 7 \end{bmatrix}$

$y = \begin{bmatrix} 3 & i & 5i \end{bmatrix}$

$(x, y) =$

$(y, x) =$

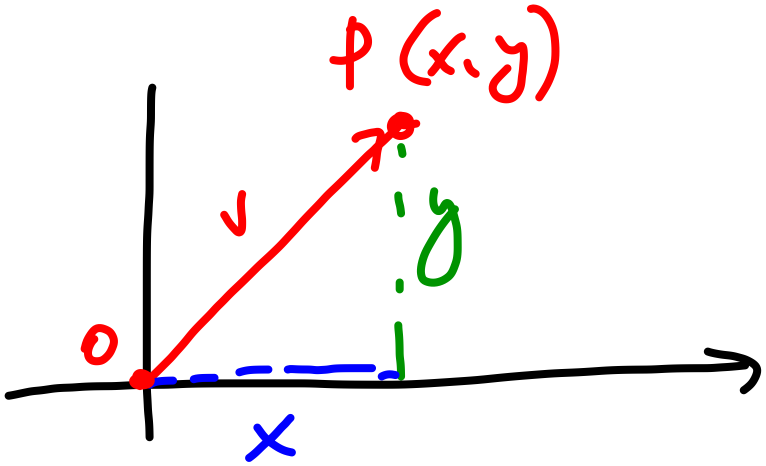
Hvis $x \in \mathbb{R}^n$ og $y \in \mathbb{R}^n$, så $x \cdot y = y \cdot x$

Egenskaper i $\mathbb{R}^n$

* $x \cdot y = y \cdot x$
* $(x + y) \cdot z = x \cdot z + y \cdot z$
* $(ax) \cdot y = a(x \cdot y)$
* $x \cdot x \geq 0$ og $x \cdot x = 0 \iff x = \vec{0}$

$\mathbb{C}^n$?

Lengde til vektorer



$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

$|OP| = \sqrt{x^2+y^2}$ (Pyth.)

<u>Def</u> lengden til en vektor $\vec{v}$ i $\mathbb{R}^n$ $(\mathbb{C}^n)$ er gitt ved

$$|\vec{v}| = \sqrt{(\vec{v},\vec{v})} = \sqrt{v_1^2+v_2^2+\dots+v_n^2}$$

Ex. $\vec{v} = [2, i, 7]$

$|\vec{v}| =$

Ex 2 $\vec{v} = [1\ 0\ 0\ 0]$

$|\vec{v}| =$

Def en vektor $\vec{u}$ med lengde $|u| = 1$
kalles en enhetsvektor

Normalisering (retning)

$\vec{v} \neq \vec{0} \Rightarrow \vec{u} = \frac{1}{|\vec{v}|}\vec{v}$. $\quad |\vec{u}| = 1$

Vektornormer generelt

V. norm i $\mathbb{C}^n$ er en funksjon med reelle verdier slik at

1) $\|\vec{x}\| \geq 0 \quad \forall \vec{x} \in \mathbb{C}^n$

   $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$

2) $\|\alpha x\| = |\alpha| \, \|x\| \quad \forall x \in \mathbb{C}^n, \forall \alpha \in \mathbb{C}$

3) $\|x + y\| \leq \|x\| + \|y\| \quad \forall x, y \in \mathbb{C}^n$

Hvis det finns indreprodukt, så 2-norm er

$$\|\bar{x}\|_2 = \sqrt{(\bar{x}, \bar{x})}$$

Hölder p-normer

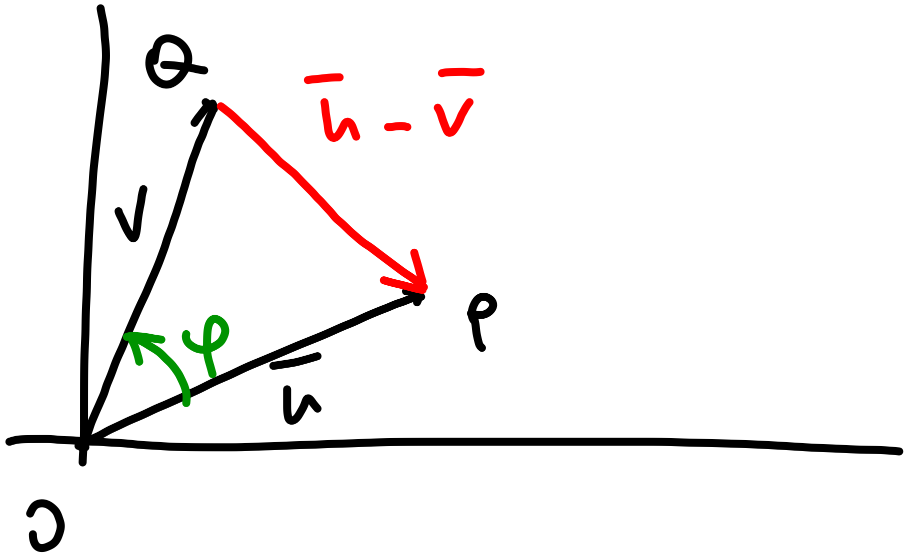
$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} , \quad p \geq 1$$

$$\|x\|_1 = |x_1| + |x_2| + .. + |x_n|$$

$$\|x\|_2 = \left(x_1^2 + x_2^2 + .. + x_n^2 \right)^{1/2}$$

$$\|x\|_\infty = \max_{i=1,..,n} |x_i|$$

Norm og avstand



avstand mellom P og Q

er $|\bar{u}-\bar{v}|$

Def: $\vec{u} \in \mathbb{C}^n$, $\vec{v} \in \mathbb{C}^n$

$$d(\vec{u},\vec{v}) = |\vec{u}-\vec{v}| = \|\vec{u}-\vec{v}\|_2$$

$$|\bar{u}-\bar{v}|^2 = |\bar{u}|^2 + |\bar{v}|^2 - 2|\bar{u}||\bar{v}|\cos\varphi$$

$$\cos\varphi = \frac{\boxed{}(\vec{u},\vec{v})}{|u|\cdot|v|}$$

: vinkel mellom vektorene

Omformulering

$$(\vec{a}, \vec{b}) = |\vec{a}|\,|\vec{b}|\,\cos\varphi \quad \| \; \mathbb{R}^2$$

$0 \qquad \neq 0 \qquad \neq 0$

$$\cos\varphi = 0$$

$$\varphi = \frac{\pi}{2}$$

$$|\vec{a}|\cos\varphi = \frac{(\vec{a}, \vec{b})}{|\vec{b}|}$$

Spesialtilfelle. orthogonale vektorer

$\vec{u} \in \mathbb{C}^n$, $\vec{v} \in \mathbb{C}^n$

$\vec{u}$ og $\vec{v}$ er orthogonale hvis $\vec{u} \cdot \vec{v} = 0$

Et sett $G = \{a_1, a_2, \dots, a_n\}$ er orthogonalt

hvis $(a_i, a_j) = 0$ hvis $i \neq j$.

Et sett G er orthonormalt hvis

- det er orthogonal
- $\|a_i\|_2 = 1 \quad \forall i = 1, \dots, n$

Ex

$$v_1 = [1 \ \ 0 \ \ 2 \ \ -i]$$

$$v_2 = [-2 \ \ 2 \ \ 1 \ \ 0]$$

$$v_3 = [0 \ \ 1 \ \ 3 \ \ -4i]$$

$$v_3'' = [0, 1, -2, -4i]$$

Knallegenskap:

<u>th</u> Et orthogonalt vektorsett er lineært uavhengig

Direkte sum

S_1, S_2 - underrom

Sum av S_1 og S_2: $\{v = x_1 + x_2 \mid x_1 \in S_1, x_2 \in S_2\}$

Hvis $S_1 \cap S_2 = \{\vec{0}\}$, så skriver vi

$$S = S_1 \oplus S_2$$

Hvis $S = \mathbb{C}^m$, så $\forall x \in \mathbb{C}^m$ kan skrives som

$$x = x_1 + x_2, \quad x_1 \in S_1, \quad x_2 \in S_2$$

og representasjonen er unik

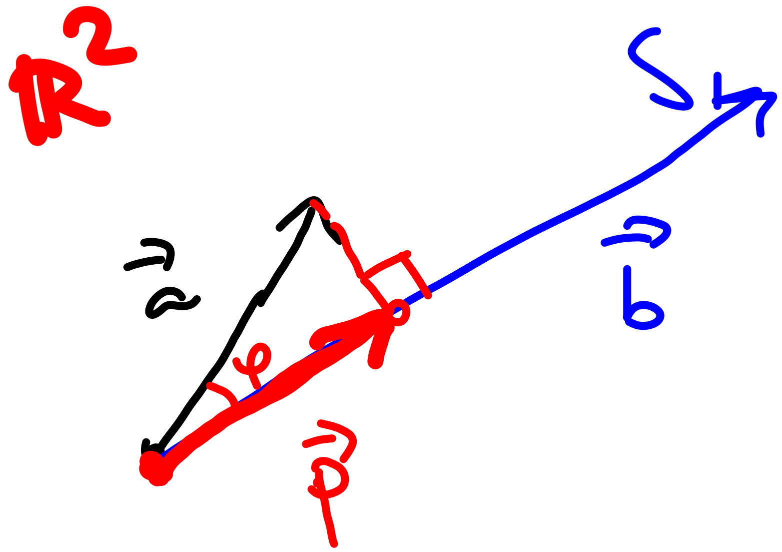
$$\dim S_1 + \dim S_2 =$$

$P \quad x \to x_1$ kalles for projektor på S_1 langs S_2.

Projeksjoner

$\mathbb{C}^n = S_1 \oplus S_2$ $\quad \forall \bar{x} \in \mathbb{C}^n \quad \bar{x} = \underset{\in S_1}{\bar{x}_1} + \underset{\in S_2}{\bar{x}_2}$

$P: \bar{x} \to \bar{x}_1$ projeksjon ned på S_1



vektor retning, lengde

1) retning
2) lengde

Projeksjoner 2 $\mathbb{R}^n$

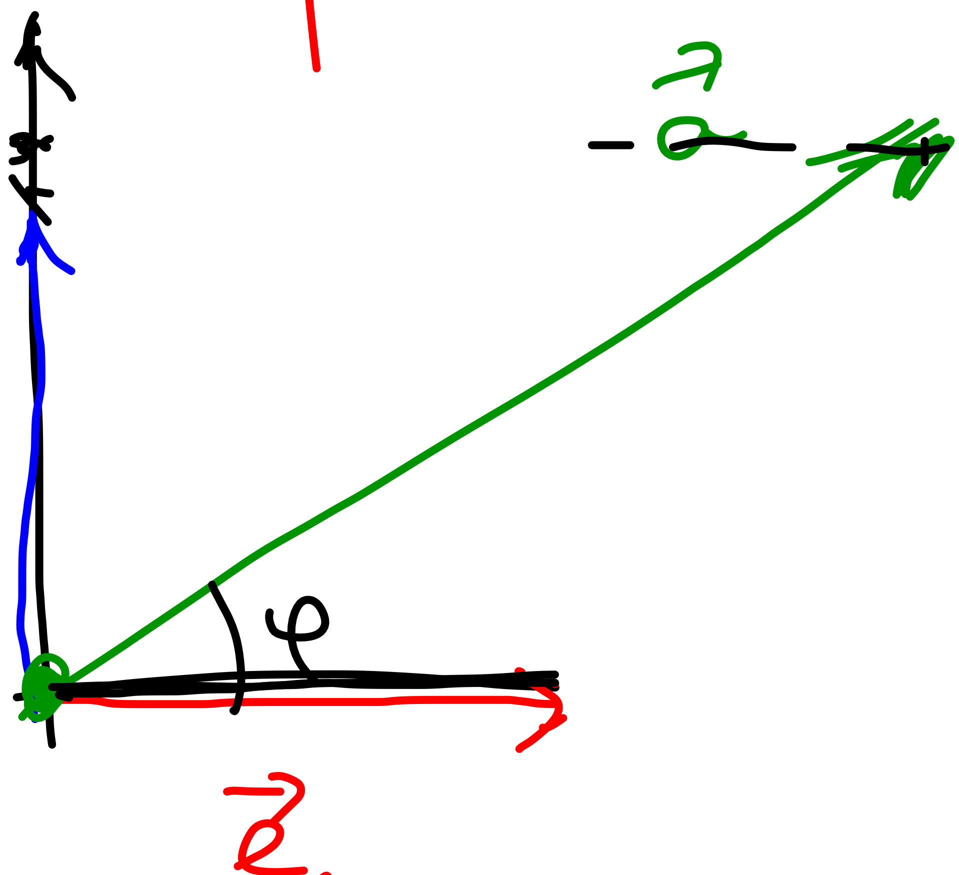
projeksjon $\vec{p} = |\vec{p}|\,\vec{r} = \frac{(\vec{a},\vec{b})}{|\vec{b}|^2}\,\vec{b} =$

$=$

$= \frac{(\vec{a},\vec{b})}{(\vec{b},\vec{b})} \cdot \vec{b} =$

ortogonal projeksjon av $\vec{a}$ ned på $\vec{b}$

$\mathbb{R}^2 = \operatorname{span}\{e_1\} \oplus \operatorname{span}\{e_2\}$



$|\vec{a}| \cdot \cos\varphi = |\vec{p}_{e_1}|$

$$|\vec{p}_{e_1}| = \frac{(\vec{a} \cdot \vec{e}_1)}{|e_1|}$$

$$\vec{p}_{e_2} = \frac{(\vec{a} \cdot \vec{e}_2)}{|\vec{e}_2|} \cdot \vec{e}_2$$

Ex

$W = \text{span}\{\underbrace{(1,2,0,0)^T}_{\vec{v}_1}\}$

$\vec{y} = (1 \;\; 2 \;\; 3 \;\; 4)^T$

finn projeksjonen av $\vec{y}$ ned på W

$$\text{proj}_W(y) = \frac{(\vec{v}_1, \vec{y})}{(\vec{v}_1, \vec{v}_1)} \cdot \vec{v}_1 =$$

$(v_1, v_1) =$

$(v_1, y) =$

Hvis vi kjenner til ortogonal basis $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k\}$ i W, så kan vi generalisere

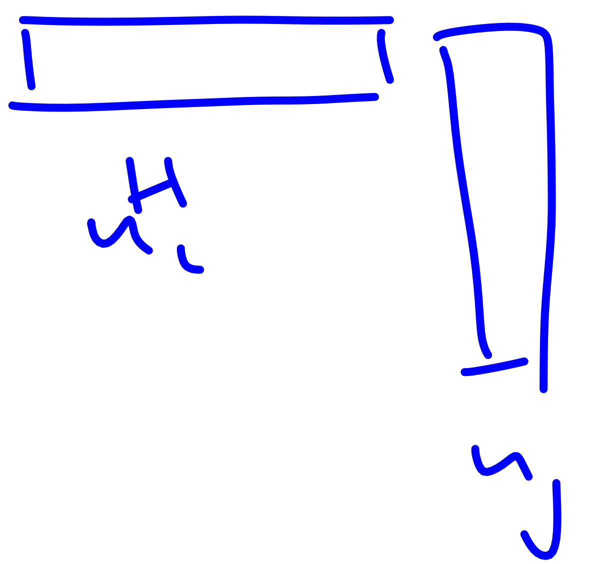
$$\vec{p} = \frac{(\vec{a}, \vec{b})}{(\vec{b}, \vec{b})} \vec{b} = \mathrm{proj}_{\vec{b}}(\vec{a})$$

$$\mathrm{proj}_W(\vec{a}) = \frac{(\vec{a}, \vec{u}_1)}{(\vec{u}_1, \vec{u}_1)} \vec{u}_1 + \ldots + \frac{(\vec{a}, \vec{u}_k)}{(\vec{u}_k, \vec{u}_k)} \vec{u}_k$$

Ortonormale vektorsett $\{u_1, u_2, \ldots, u_k\}$

- ortogonale
- alle vektorene har lengde 1

dvs $$u_i^H u_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$



La oss velge et ortonormalt basis i $\mathbb{C}^n$

$V = \{\bar{u}_1, \quad , \bar{u}_n\}$

Så lager vi ei matrise

$$\mathcal{U} = [\bar{u}_1 | \bar{u}_2 | \ldots | \bar{u}_n]$$

$$(\mathcal{U}^H \mathcal{U})_{ij} = (\bar{u}_j^H \; \bar{u}_i) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

m.a.o. $\quad \mathcal{U}^H \mathcal{U} = I = \mathcal{U}\mathcal{U}^H, \quad \mathcal{U}^H = \mathcal{U}^{-1}$

unitary / unitære

Egenskaper av unitære matriser

1) $|U\bar{x}| = |\bar{x}|$

2) $(U\bar{x})\cdot(U\bar{y}) = \bar{x}\cdot\bar{y}$

$$(U\bar{x})\cdot(U\bar{y}) = (U\bar{y})^H U\bar{x} = \bar{y}^H U^H U\bar{x} =$$

$$= \bar{y}^H I\bar{x} = \bar{y}^H\bar{x} = (\bar{x}\cdot\bar{y})$$

Gram-Schmidt: hvordan lage et orthonormalt sett

GH $G: \{x_1, x_2, x_3, \ldots x_n\}$ et sett med vekt

Regner ut $Q: \{q_1, q_2, \ldots, q_n\}$ et orthonormalt sett

$$q_1 = \frac{1}{\|x_1\|_2} \cdot x_1$$

$$\hat{q}_2 = x_2 - \underbrace{(x_2, q_1) \cdot q_1}_{\text{projeksjon}} \qquad q_2 = \frac{1}{\|\hat{q}_2\|_2} \cdot \hat{q}_2$$

$$\hat{q}_3 = x_3 -$$

Alg <u>GS</u>

1 Start: $r_{11} = \|x_1\|_2$ if $r_{11} = 0$ stop
elseif $q_1 = x_1 / r_{11}$

2 Loop

for $j = 2..n$ do

for $i = 1..j-1$ do

$r_{ij} = (x_j, q_i)$

endfor

$\hat{q} = x_j - \sum_{i=1}^{j-1} r_{ij} q_i$

$r_{jj} = \|\hat{q}\|_2$ if $r_{jj} = 0$ stop elseif $q_j = \hat{q} / r_{jj}$

endfor

$$x_j = \sum_{i=1}^{j} r_{ij} q_i$$

$$X = [x_1 | x_2 | . | x_n]$$

$$R = (r_{ij})$$

$$Q = [q_1 | q_2 | \dots | q_n]$$

$$\Rightarrow \quad X = QR$$

QR-faktorisering

Compare the results of classical Gram–Schmidt and modified Gram–Schmidt, computed in double precision, on the matrix of almost-parallel vectors

$$\begin{array}{c} \begin{bmatrix} 1 & \boxed{\begin{matrix} 1 \\ 0 \\ \delta \\ 0 \end{matrix}} & 1 \\ \delta & & 0 \\ 0 & & 0 \\ 0 & & \delta \end{bmatrix} \\ A_1 \quad A_2 \quad A_3 \end{array}$$

$(q_1 ; q_2) = \delta \left(-\frac{1}{\sqrt{2}}\right) =$

$= c \cdot \delta$

where $\delta = 10^{-10}$.

First, we apply classical Gram–Schmidt.

$$y_1 = A_1 = \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} \qquad \text{and} \qquad q_1 = \frac{1}{\sqrt{1+\delta^2}} \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix}.$$

Note that $\delta^2 = 10^{-20}$ is a perfectly acceptable double precision number, but $1 + \delta^2 = 1$ after rounding. Then

$$y_2 = \begin{bmatrix} 1 \\ 0 \\ \delta \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} q_1^T A_2 = \begin{bmatrix} 1 \\ 0 \\ \delta \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\delta \\ \delta \\ 0 \end{bmatrix} \quad \text{and} \quad q_2 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

after dividing by $||y_2||_2 = \sqrt{\delta^2 + \delta^2} = \sqrt{2}\delta$. Completing classical Gram–Schmidt,

$$y_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \delta \end{bmatrix} - \underbrace{\begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix}}_{q_1} q_1^T A_3 - \cancel{\underbrace{\begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}}_{q_2} q_2^T A_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \delta \end{bmatrix} - \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\delta \\ 0 \\ \delta \end{bmatrix} \quad \text{and} \quad q_3 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Unfortunately, due to the double precision rounding done in the first step, q_2 and q_3 turn out to be not orthogonal:

$$q_2^T q_3 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}^T \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{2}.$$

GS' (uten normalisering)

q_1

$q_2 = x_2 - (x_2, q_1) q_1$

$q_3 = x_3 - (x_3, q_1) q_1 - (x_3, q_2) q_2$

$q_4 = x_4 - (x_4, q_1) q_1 - (x_4, q_2) q_2 - (x_4, q_3) q_3$

MGS

q_1 er ferdig

$q_2 = x_2 - (x_2, q_1) q_1$

$q_3 = x_3 - (x_3, q_1) q_1$

$\vdots$

$q_k = x_k - (x_k, q_1) q_1$

q_2 er ferdig

$q_3 = q_3 - (q_3, q_2) q_2 = x_3 - (x_3, q_1) q_1 - (q_3, q_2)$

$\vdots$

$q_k = q_k - (q_k, q_2) q_2$

og så videre

q_3 er ferdig

...

$$(\underbrace{q_3}_{}; q_2) \overset{\vec{q}_2}{=} (\underbrace{x_3 - (x_3, q_1) q_1}_{}; q_2) =$$

$$= (x_3; q_2) - ((x_3; q_1) q_1; q_2) =$$

$$= (x_3; q_2) - (x_3; q_1) \underbrace{(q_1; q_2)}_{= \cancel{0} \quad 10^{-10}} =$$

$$= (x_3, q_2)$$

Alg GS-mod

1 Start: $r_{11} = \|x_1\|_2$ if $r_{11} = 0$ stop elseif $q_1 = x_1 / r_{11}$

2 Loop

gammel

for $j = 2 .. n$ do

for $i = 1 .. j-1$ do

$r_{ij} = (x_j; q_i)$

endfor

$\hat{q} = x_j - \sum_{i=1}^{j-1} r_{ij} q_i$

ny

$\hat{q} = x_j$

for $i = 1 .. j-1$ do

$r_{ij} = (\hat{q}, q_i)$

$\hat{q} = \hat{q} - r_{ij} q_i$

endfor

$r_{jj} = \|\hat{q}\|_2$

$\rightarrow$ if $r_{jj} = 0$ stop elseif $q_j = \hat{q} / r_{jj}$

endfor

On the other hand, modified Gram–Schmidt does much better. While q_1 and q_2 are calculated the same way, q_3 is found as

$$q_2 = \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$y_3^1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \delta \end{bmatrix} - \begin{bmatrix} 1 \\ \delta \\ 0 \\ 0 \end{bmatrix} q_1^T A_3 = \begin{bmatrix} 0 \\ -\delta \\ 0 \\ \delta \end{bmatrix},$$

$$y_3 = y_3^1 - \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} q_2^T y_3^1 = \begin{bmatrix} 0 \\ -\delta \\ 0 \\ \delta \end{bmatrix} - \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \frac{\delta}{\sqrt{2}}$$

$$= \begin{bmatrix} 0 \\ -\frac{\delta}{2} \\ -\frac{\delta}{2} \\ \delta \end{bmatrix} \quad \text{and} \quad q_3 = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}.$$

$$(q_2; q_3) = -\frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{6}}\right) + \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{6}}\right) = 0$$